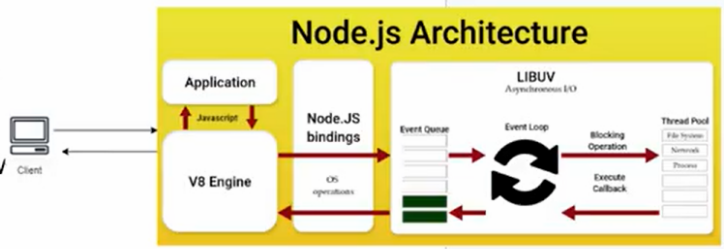
Link:

https://www.youtube.com/watch?v=GpzabO_fXUk&list=PLF7ny2pVBcm1RZVXmYSDr1gGR0cJJQlIT&index=1

- What is nodejs & How it works internally
- Software installation
- First programe in Nodejs
- Explore file system
- What are events and how



- Modules
- Promise & it's types
- Async await
- What is REST architecture
- Introduction to express framwork
  - How routing works
  - CRUP operations of /products api with static data

- What is express middleware
  - What is custom middleware
  - Validate req body
  - Validate jwt token

- Introduction to Atlas MongoDB.
- Develop Authrization API with NodeJs,Express and MongoDB

Node.js is an **open-source** and **cross-platform** JavaScript **runtime environment**.

Node.js provides a runtime environment outside of the browser. It's also built on the Chrome V8 JavaScript engine. This makes it possible to build back-end applications using the same JavaScript

Nodejs built for Highly-scalable, data-intensive and real time apps

Why Nodejs

Javascript

Community support

Rapid development

Highly scalable and fast server

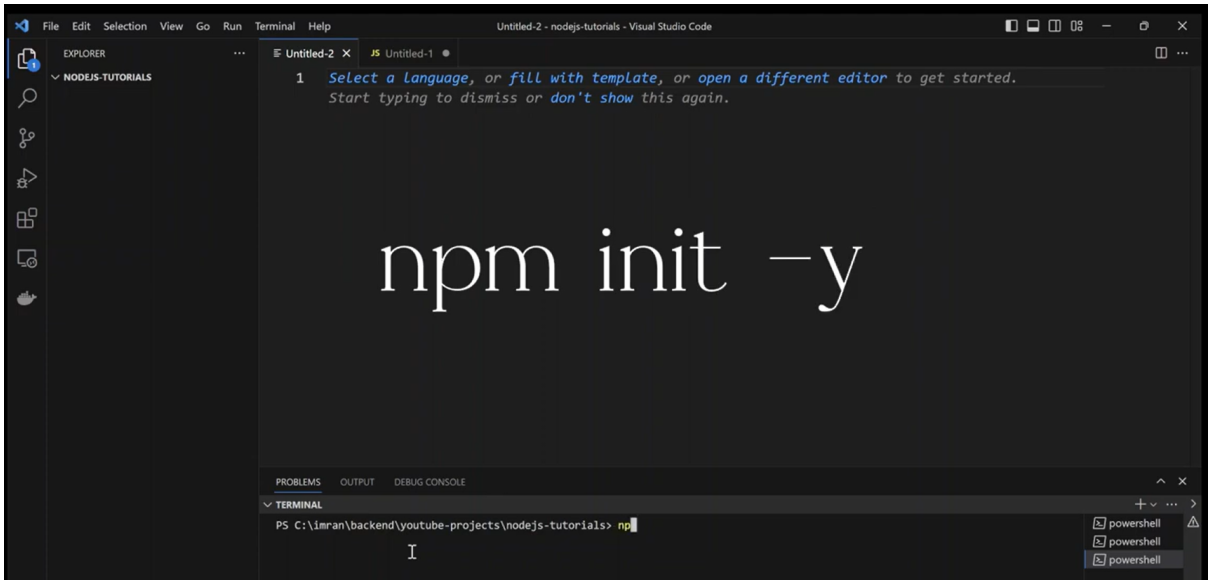
Learn How to install Node.js, NVM (Node Version Manager) and visual studio code.
Software Installation Links
Nodejs - https://nodejs.org/en
NVM - https://github.com/coreybutler/nvm-wi...
Visual Studio Code- https://code.visualstudio.com/

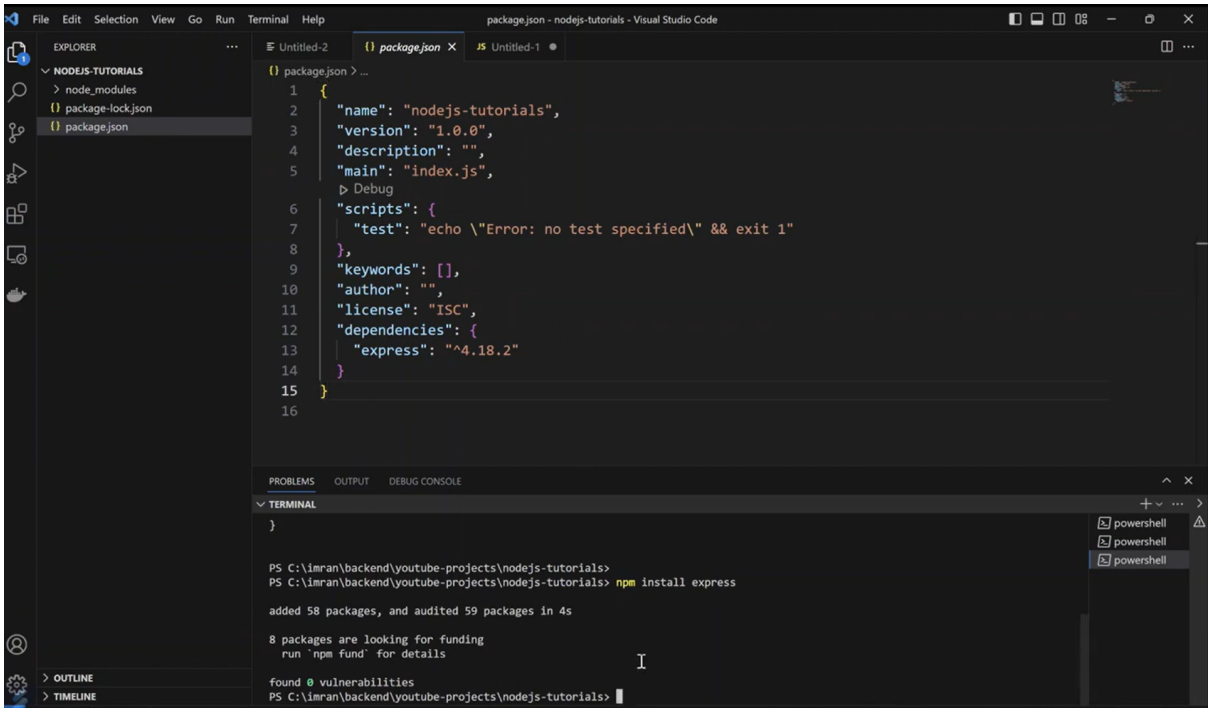
If we requires different versions of NodeJS for different project, we have to install NVM (Node version Manager) commands given below.

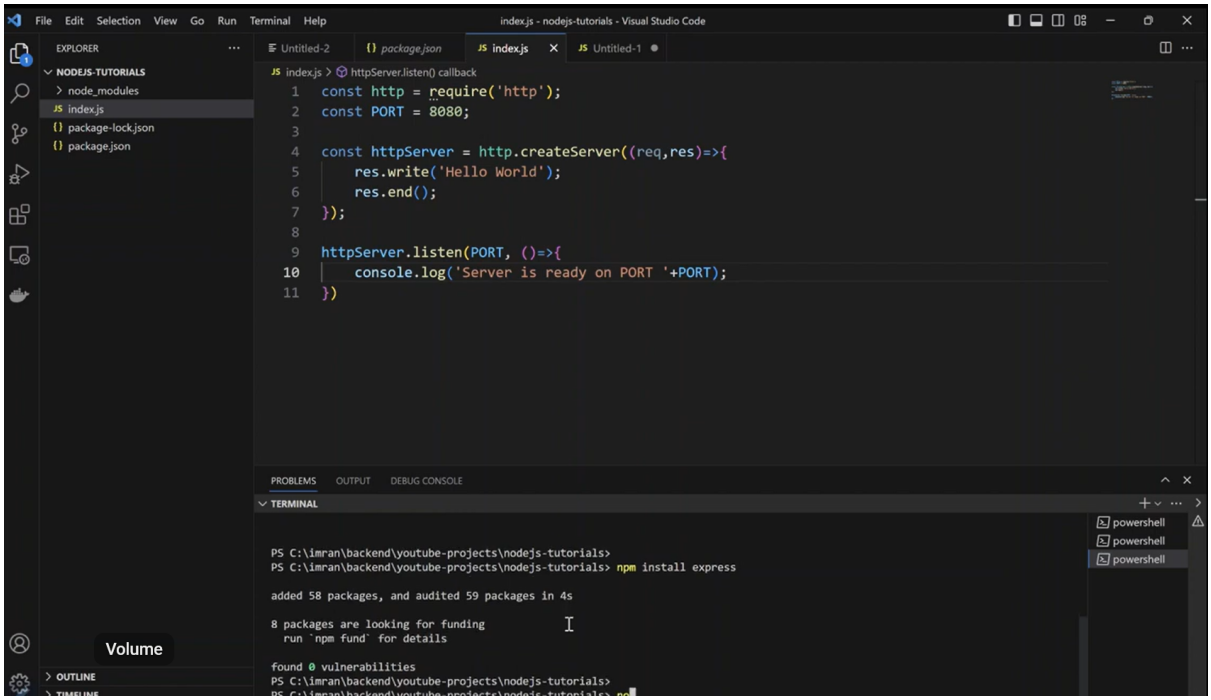
```
C:\Users\91877>nvm -v
1.1.11

C:\Users\91877>nvm install latest
21.1.0
Downloading node.js version 21.1.0 (64-bit)...
Extracting node and npm...
Complete
npm v10.2.0 installed successfully.


Installation complete. If you want to use this version, type

nvm use 21.1.0

C:\Users\91877>nvm list

    21.1.0

C:\Users\91877>node -v
'node' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\91877>nvm use 21.1.0
Now using node v21.1.0 (64-bit)

C:\Users\91877>node -v
v21.1.0
```

```
C:\Users\91877>node -v
v21.1.0

C:\Users\91877>
C:\Users\91877>
C:\Users\91877>nvm install 18.17.1
Downloading node.js version 18.17.1 (64-bit)...
Extracting node and npm...
Complete
npm v9.6.7 installed successfully.


Installation complete. If you want to use this version, type

nvm use 18.17.1

C:\Users\91877>nvm list

  * 21.1.0 (Currently using 64-bit executable)
    18.17.1

C:\Users\91877>nvm use 18.17.1
Now using node v18.17.1 (64-bit)

C:\Users\91877>
C:\Users\91877>
C:\Users\91877>node -v
v18.17.1
```

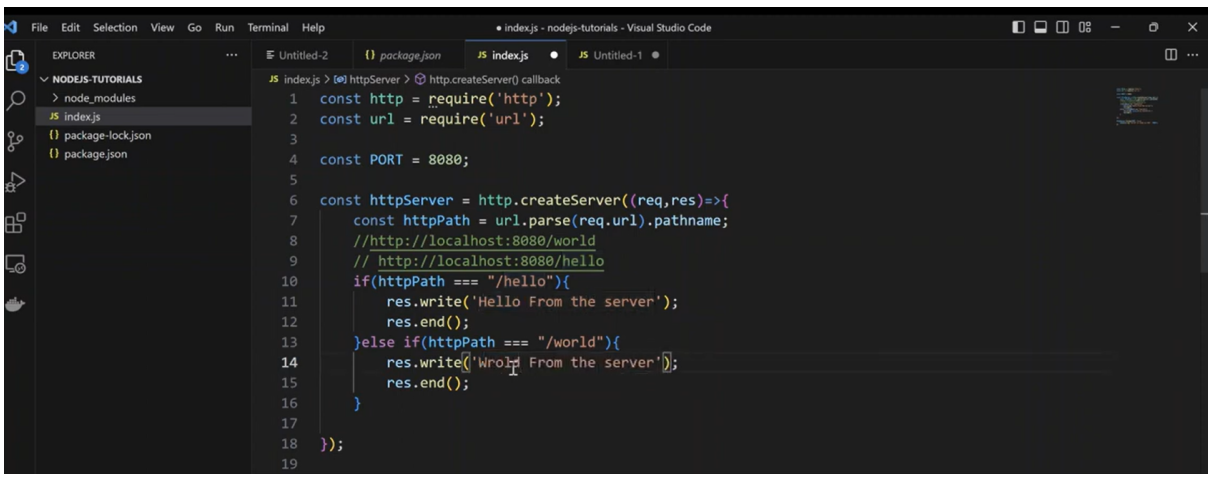
First Program of Node js


File Edit Selection View Go Run Terminal Help
Untitled-2 - nodejs-tutorials - Visual Studio Code
EXPLORER
NODEJS-TUTORIALS
Untitled-2
Untitled-1
1 Select a language, or fill with template, or open a different editor to get started.
Start typing to dismiss or don't show this again.
npm init -y
PROBLEMS OUTPUT DEBUG CONSOLE
TERMINAL
PS C:\imran\backend\youtube-projects\nodejs-tutorials> np
powershell
powershell
powershell


File Edit Selection View Go Run Terminal Help
package.json - nodejs-tutorials - Visual Studio Code
EXPLORER
NODEJS-TUTORIALS
node_modules
package-lock.json
package.json
Untitled-2
package.json
Untitled-1
{} package.json > ...
{
"name": "nodejs-tutorials",
"version": "1.0.0",
"description": "",
"main": "index.js",
Debug
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"keywords": [],
"author": "",
"license": "ISC",
"dependencies": {
"express": "^4.18.2"
}
}
PROBLEMS OUTPUT DEBUG CONSOLE
TERMINAL
}
PS C:\imran\backend\youtube-projects\nodejs-tutorials>
PS C:\imran\backend\youtube-projects\nodejs-tutorials> npm install express
added 58 packages, and audited 59 packages in 4s
8 packages are looking for funding
run `npm fund` for details
found 0 vulnerabilities
PS C:\imran\backend\youtube-projects\nodejs-tutorials>
powershell
powershell
powershell
OUTLINE
TIMELINE



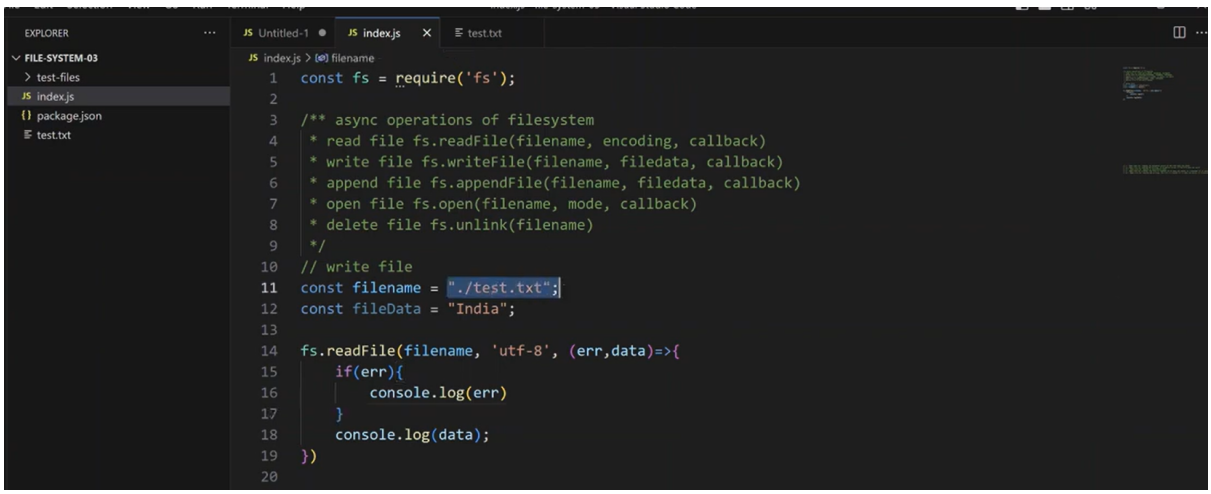
Dynamic routing

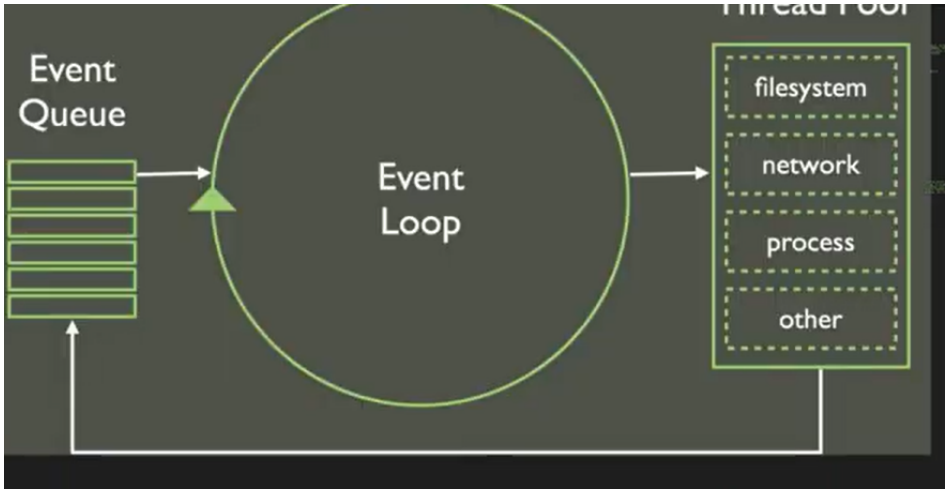


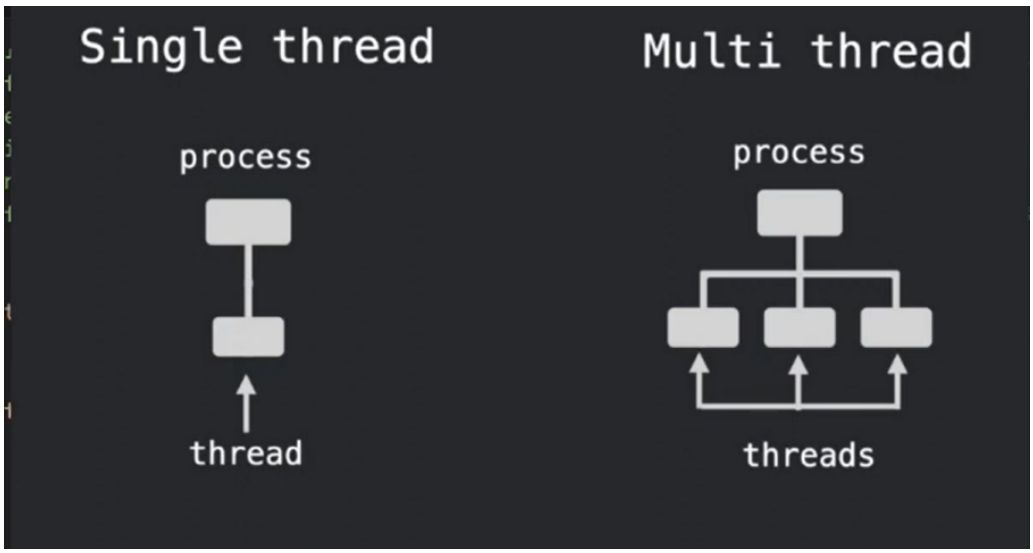
File System in Nodejs

- **Async Operations**
  Read, Write, Open
- **Sync Operations**
  Read, Write, Open
- **Promise Based Operations**
  Read, Write, Open

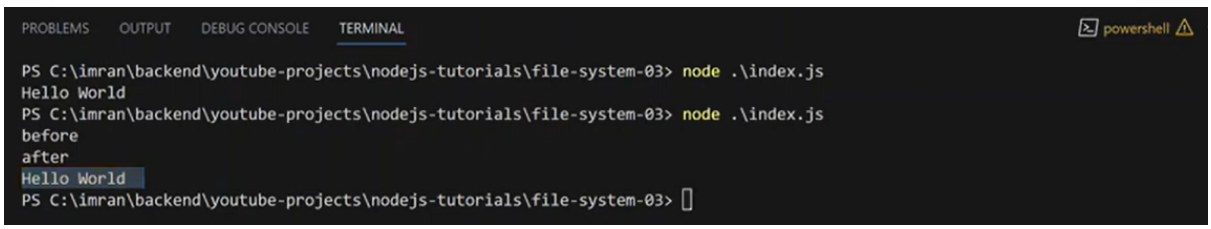
Async operation

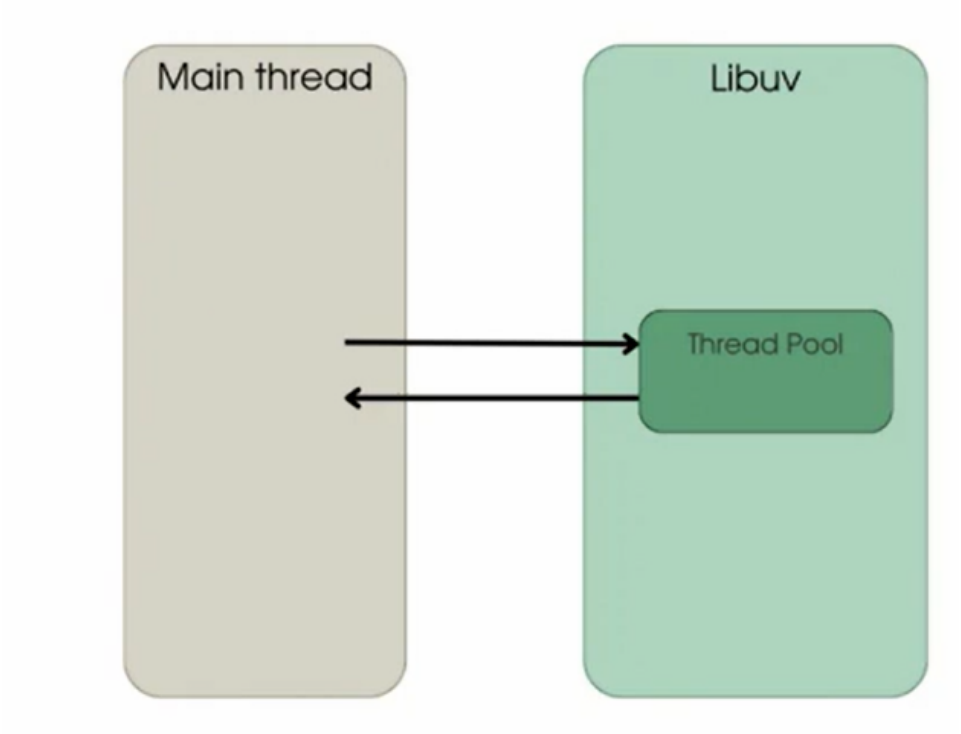



Event
Queue
Event
Loop
filesystem
network
process
other


Single thread
Multi thread
process
process
thread
threads

Output


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
powershell
PS C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03> node .\index.js
Hello World
PS C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03> node .\index.js
before
after
Hello World
PS C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03>



Write file:

```
JS Untitled-1 ●   JS index.js ×   test.txt
JS index.js > ...
 8   * delete file fs.unlink(filename)
 9   */
10  // write file
11  const filename = "./test.txt";
12  const fileData = "India";
13
14  fs.writeFile(filename, fileData ,(err,result)=>{
15      if(err){
16          console.log(err)
17      }
18      console.log('File write success')
19  })
20
21  //single thread
22
```

Append file

```
11  const filename = "./test.txt";
12  const fileData = "India";
13
14  fs.appendFile(filename, fileData ,(err,result)=>{
15      if(err){
16          console.log(err)
17      }
18      console.log('File A success')
19  })
20
21  //single thread
22
```

Open file:

```
13
14  fs.open(filename, 'r', (err,fd)=>{
15      if(err){
16          console.log(err)
17      }
18      fs.readFile(fd,'utf-8',(err,data)=>{
19          if(err){
20              console.log(err)
21          }
22          console.log('Read file sucess ',data);
23      })
24  })
```

```
ROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL

fter
ile Data:  Hello World
S C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03> node .\index.js
ile write success
S C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03> node .\index.js
ile Append success
S C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03> node .\index.js
ile Append success
S C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03>
S C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03> node .\index.js
ead file sucess  IndiaIndiaIndia
S C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03>
```

```
JS index.js > ...
const fs = require('fs');
// const fsPromises = require('fs/promises');

const filename = "./test.txt";
const fileData = "\nIndia";

/** sync operations of filesystem
 * read file fs.readFileSync(filename, encoding)
 * write file fs.writeFileSync(filename, filedata)
 * append file fs.appendFileSync(filename, filedata)
 * open file fs.openSync(filename, mode)
 * delete file fs.unlinkSync(filename)
 */
// write file

```

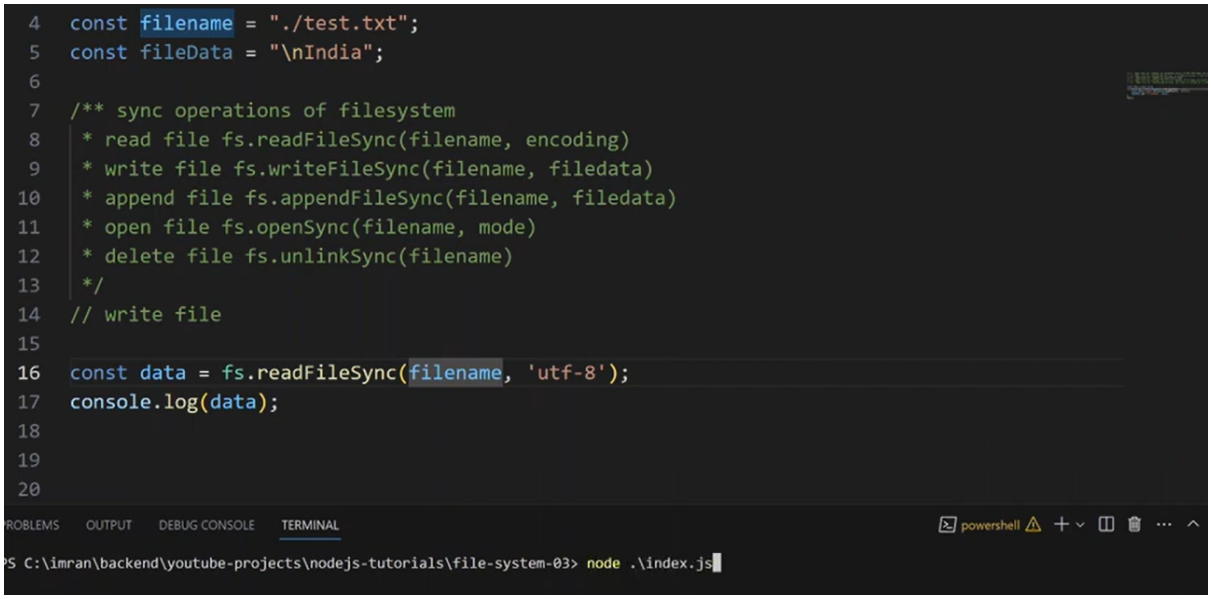
```
// r - Open file for reading, An exception occurs if the file does not exist
// r+ -Open file for reading and writing, An exception occurs if the file does not exist
// rs - Open file for reading in synchronous mode.
// w - Open file for writing. The file is created (if it does not exist) or truncated (if it ex
// w+ - Open file for reading and writing. The file is created (if it does not exist) or trunca
```

```
const filename = "./test.txt";
const fileData = "India";

fs.unlink(filename, (err,result)=>{
    if(err){
        console.log(err)
    }
    console.log('File Deleted success:')
})

//single thread

```

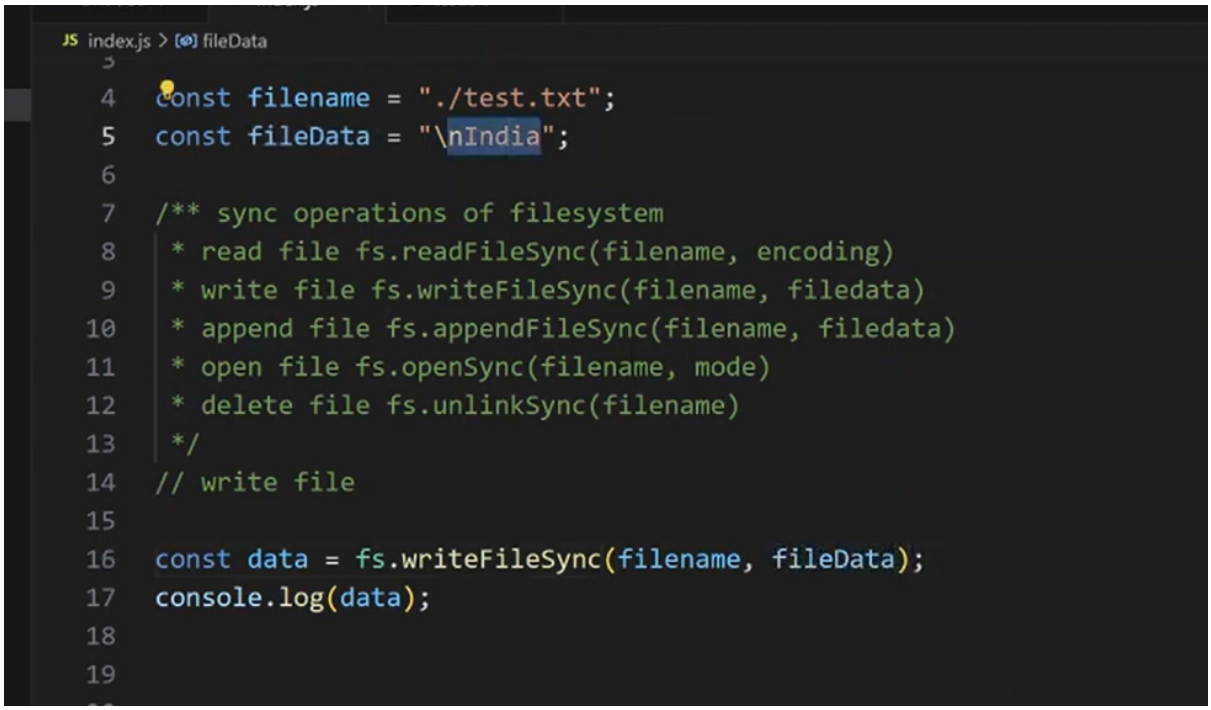
Sync operations:

```
/** sync operations of filesystem
 * read file fs.readFileSync(filename, encoding)
 * write file fs.writeFileSync(filename, filedata)
 * append file fs.appendFileSync(filename, filedata)
 * open file fs.openSync(filename, mode)
 * delete file fs.unlinkSync(filename)
 */
// write file
```

Read file:

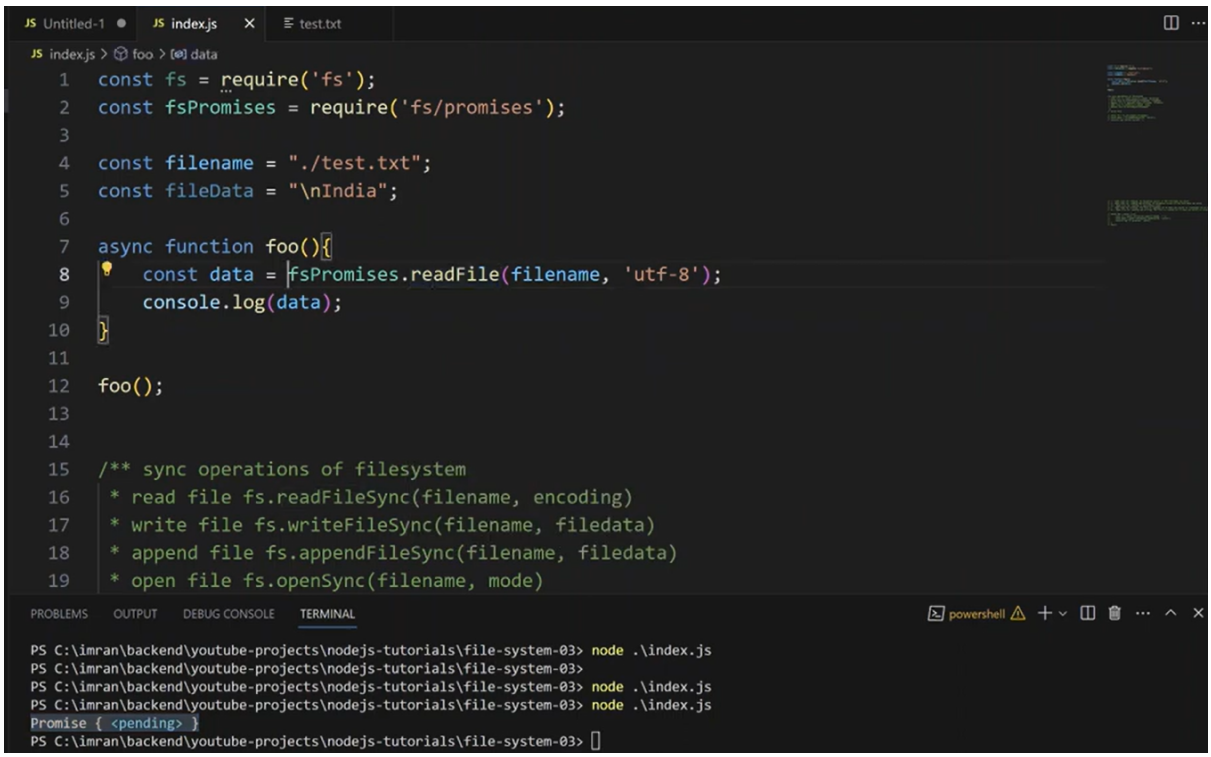

const filename = "./test.txt";
const fileData = "\nIndia";
/** sync operations of filesystem
* read file fs.readFileSync(filename, encoding)
* write file fs.writeFileSync(filename, filedata)
* append file fs.appendFileSync(filename, filedata)
* open file fs.openSync(filename, mode)
* delete file fs.unlinkSync(filename)
*/
// write file
const data = fs.readFileSync(filename, 'utf-8');
console.log(data);
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
powershell
PS C:\imran\backend\youtube-projects\nodejs-tutorials\file-system-03> node .\index.js

Write file:


index.js > fileData
const filename = "./test.txt";
const fileData = "\nIndia";
/** sync operations of filesystem
* read file fs.readFileSync(filename, encoding)
* write file fs.writeFileSync(filename, filedata)
* append file fs.appendFileSync(filename, filedata)
* open file fs.openSync(filename, mode)
* delete file fs.unlinkSync(filename)
*/
// write file
const data = fs.writeFileSync(filename, fileData);
console.log(data);

```
const filename = "./test.txt";
const fileData = "\nIndia";

/** sync operations of filesystem
 * read file fs.readFileSync(filename, encoding)
 * write file fs.writeFileSync(filename, filedata)
 * append file fs.appendFileSync(filename, filedata)
 * open file fs.openSync(filename, mode)
 * delete file fs.unlinkSync(filename)
 */
// write file

const fd = fs.openSync(filename, 'r');
const data = fs.readFileSync(fd, 'utf-8');
console.log('open in read success ',data);
```

Promises:



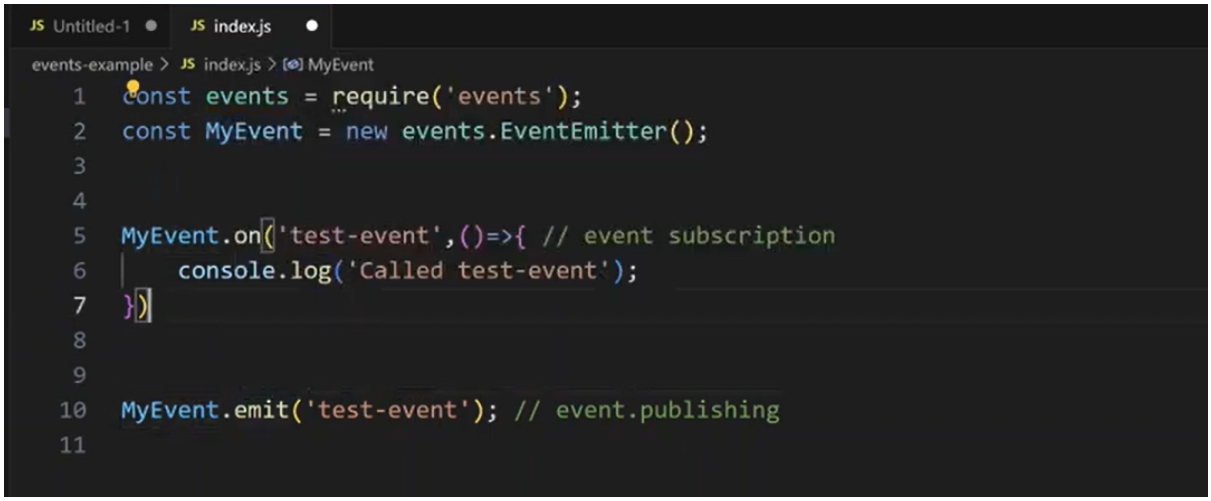
index.js > foo > data

```js
const fs = require('fs');
const fsPromises = require('fs/promises');

const filename = "./test.txt";
const fileData = "\nIndia";

async function foo(){
    try{
        const data = await fsPromises.readFile(filename, 'utf-8');
        console.log(data);
    }catch(err){
        console.log(err);
    }
}

foo();
```

```
/** sync operations of filesystem
 * read file fsPromises.readFile(filename, encoding)
 * write file fsPromises.writeFile(filename, filedata)
 * append file fs.appendFile(filename, filedata)
 * open file fs.open(filename, mode)
 * delete file fs.unlink(filename)
 */
```

Events in Nodejs

- EventEmitter
- Custom Event



Alarm smoke

Sleep alarm

Modules in Nodejs

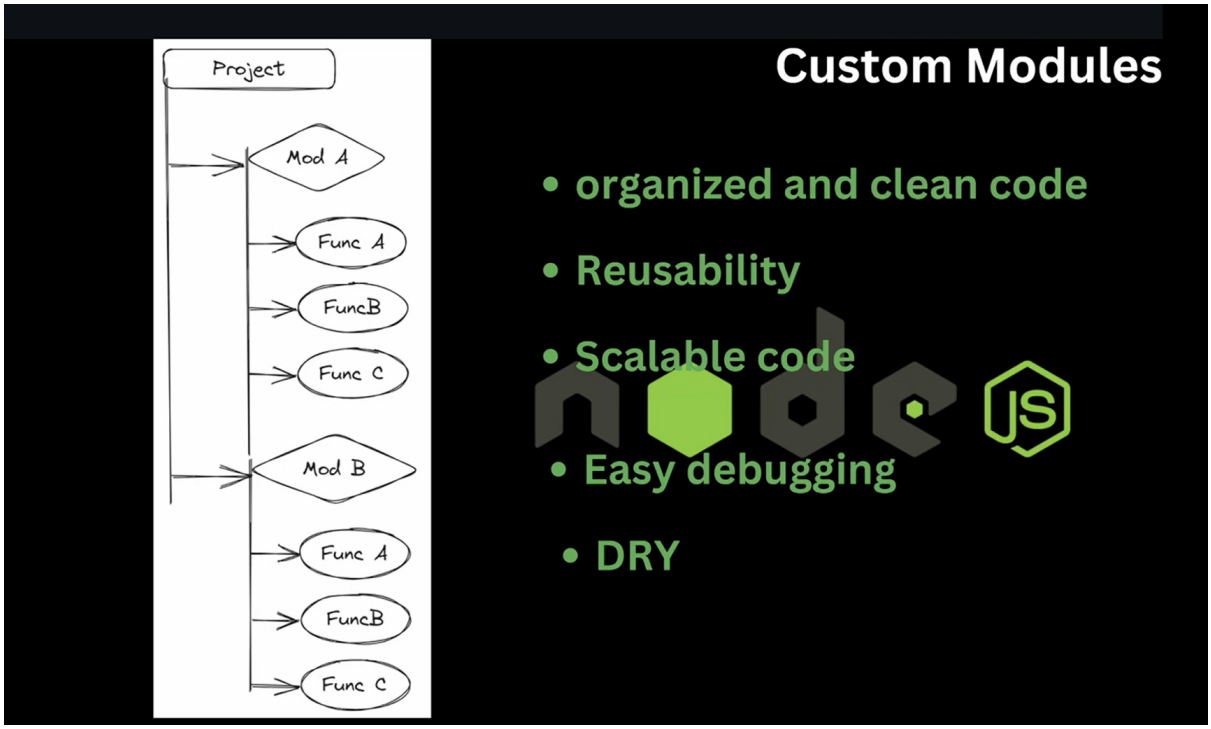
- Core Modules
- Third Party Modules
- Custom Modules

Core Modules

- fs
- HTTP
- URL
- OS
- Path

Third Party Modules

- Express
- Moment

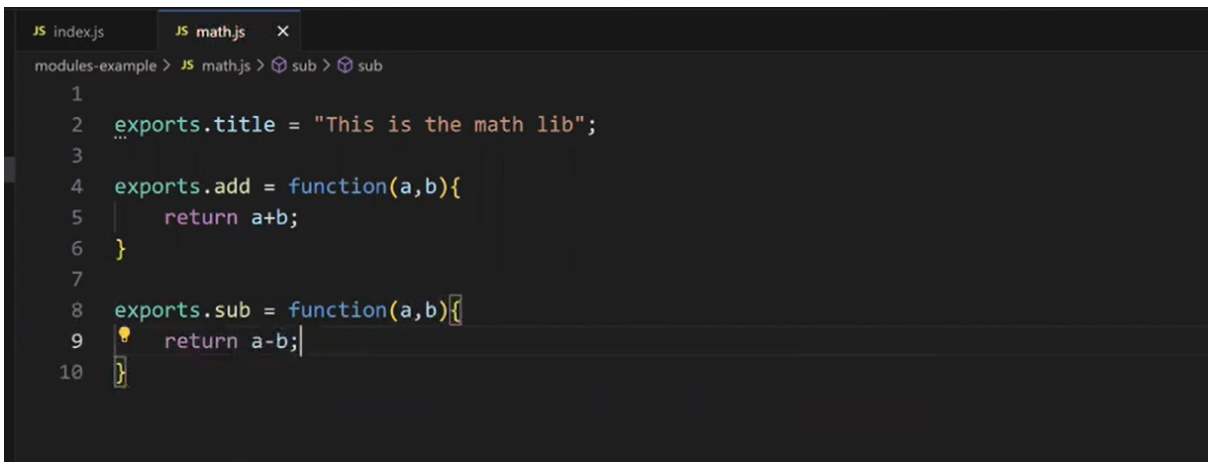


DRY: Do not repeat yourself

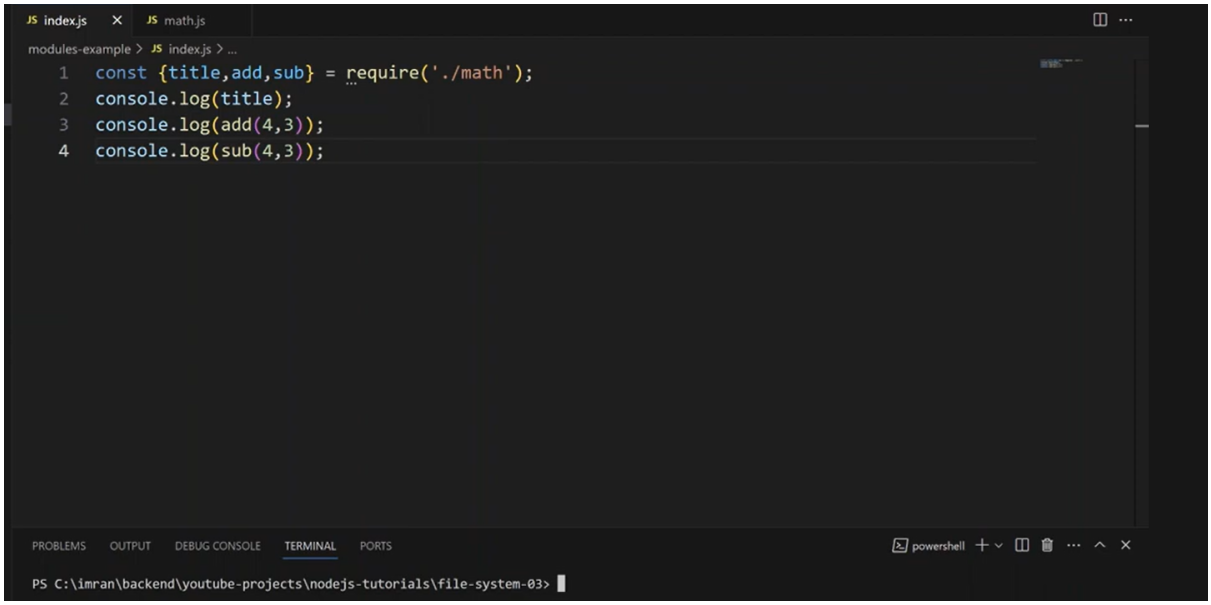
Custom Module:

Math custom Module:

Math.js

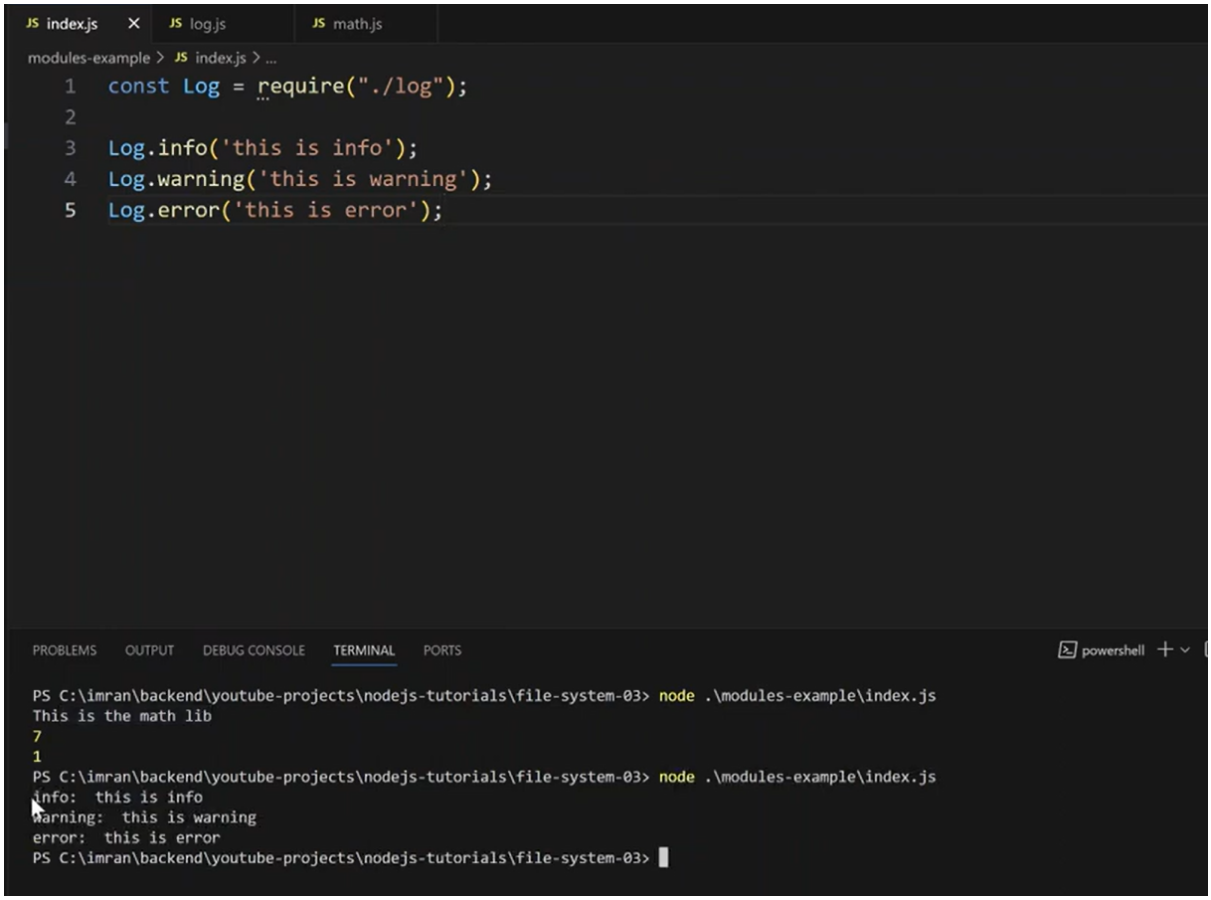


Index.js



Log.js

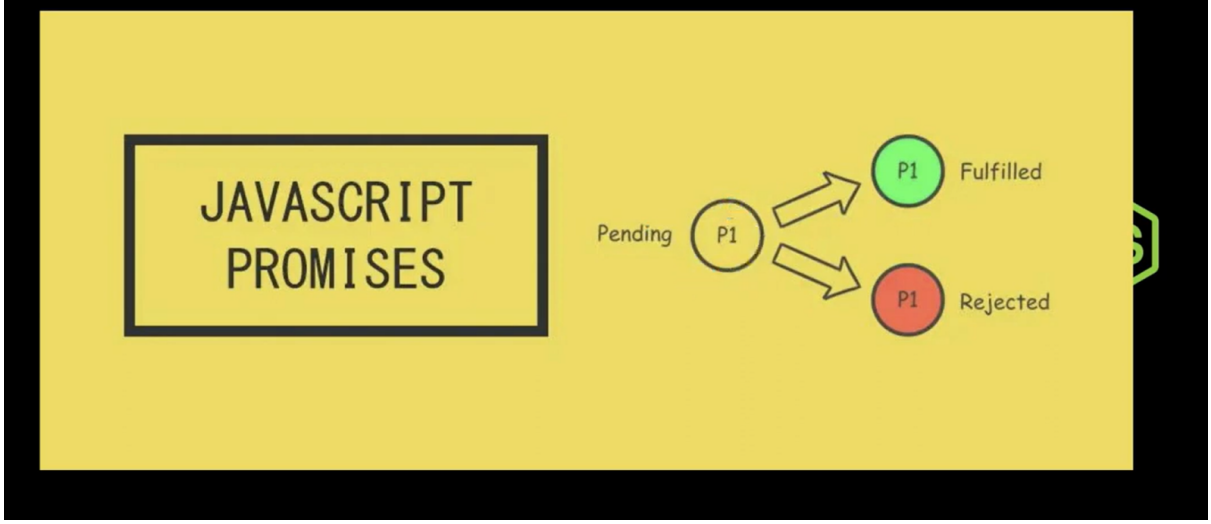
Index.js for log module



Promises

- What is Promise
- Promise states like Pending, resolved, rejected
- Promise chaining and why we need It
- Promise.all
- Promise.allSetteld
- Promise.any
- Promise.race

Promise

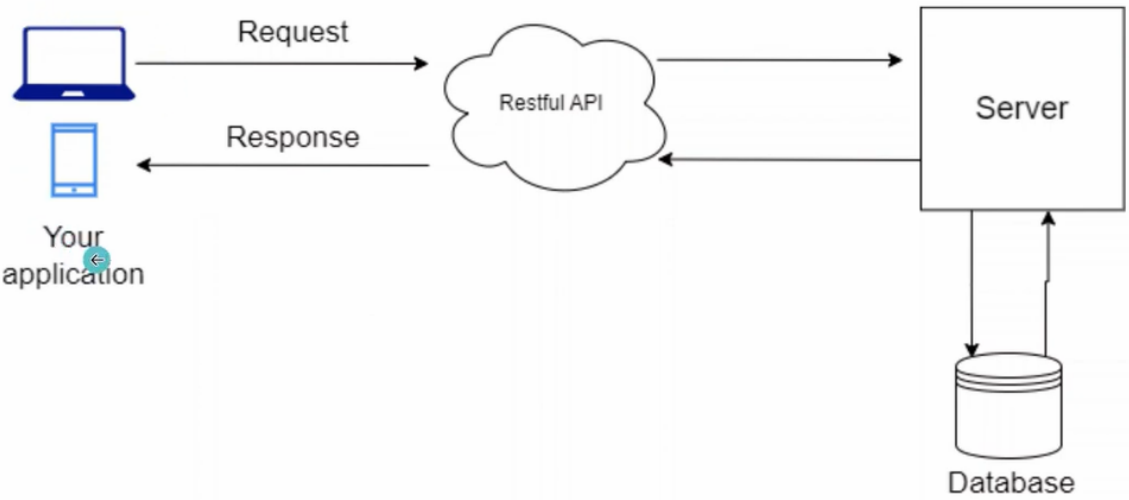



```
index.js U
index.js > p
**/

function isEligible(age){
    return new Promise((resolve,reject)=>{
        if(age<18){
            return reject('Not Eligible for Vote');
        }
        resolve('Eligible for Vote');
    })
}

const p = isEligible(19)
    .then((result)=> console.log('Success : ',result))
    .catch((err)=> console.log('Error : ',err));

console.log(p);
```

```
PS C:\imran\backend\youtube-projects\nodejs-tutorials> node .\index.js
Error :  Not Eligible for Vote
PS C:\imran\backend\youtube-projects\nodejs-tutorials> node .\index.js
Success :  Eligible for Vote
PS C:\imran\backend\youtube-projects\nodejs-tutorials> node .\index.js
Promise { <pending> }
Success :  Eligible for Vote
PS C:\imran\backend\youtube-projects\nodejs-tutorials>
```

REST

Representational State Transfer

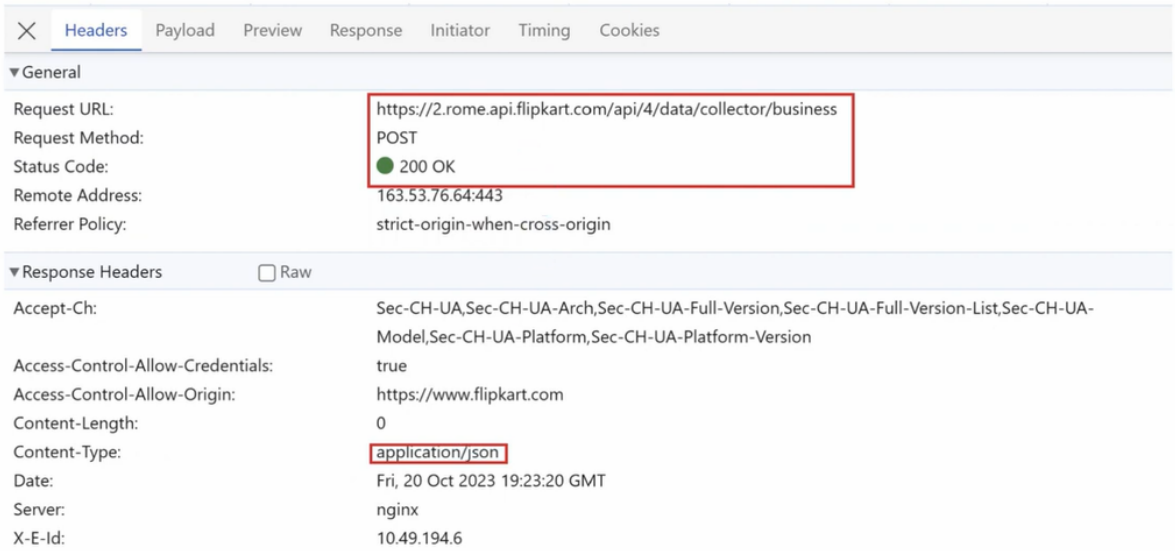


REST

- **Separation of Client and Server**
- **Statelessness**
- **Communication between Client and Server**
- **Coachability**

Make Request

- an ***HTTP verb***, which defines what kind of operation to perform
- a ***header***, which allows the client to pass along information about the request
- a ***path*** to a ***resource***
  an optional message body containing data



HTTP Verbs

GET - Fetch a specific resource (by id) or a collection of resources

POST - Create a new resource

PUT - update a specific resource (by id)

DELETE - remove a specific resource by id

Headers and Accept parameters

image — image/png, image/jpeg, image/gif

audio — audio/wav, audio/mpeg

video — video/mp4, video/ogg

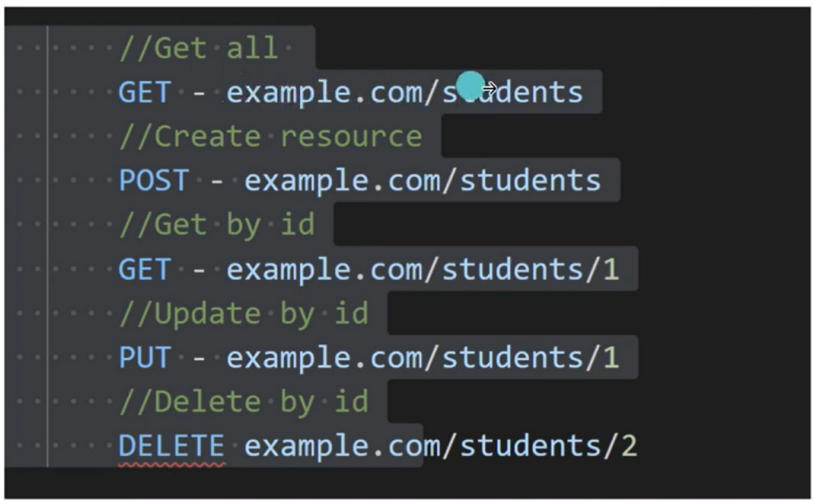
application — application/json, application/pdf, application/xml, application/octet-stream

GET /products/23
Accept: text/html, application/xhtml

Paths

paths should be self-explanatory.

Rersource path should be plural like /students



Response Codes

Status code	Meaning
200 (OK)	This is the standard response for successful HTTP requests.
201 (CREATED)	This is the standard response for an HTTP request that resulted in an item being successfully created.
204 (NO CONTENT)	This is the standard response for successful HTTP requests, where nothing is being returned in the response body.
400 (BAD REQUEST)	The request cannot be processed because of bad request syntax, excessive size, or another client error.
403 (FORBIDDEN)	The client does not have permission to access this resource.
404 (NOT FOUND)	The resource could not be found at this time. It is possible it was deleted, or does not exist yet.
500 (INTERNAL SERVER	The generic answer for an unexpected failure if there is no more

GET — return 200 (OK)
POST — return 201 (CREATED)
PUT — return 200 (OK)
DELETE — return 204 (NO CONTENT

GET all Request

```
To get all the customers

Request:
GET http://example.com/customers
Accept: application/json

Response:
Status Code: 200 (OK)
Content-type: application/JSON
```

POST Request

```
POST http://example.com/customers
Body:
{
    "name" = "Jone doe",
    "email" = "jone.doe@example.com"
}
The server then generates an ID for that object and returns it
back to the client, with a header like:

201 (CREATED)
Content-type: application/json
```

GET by Id Request

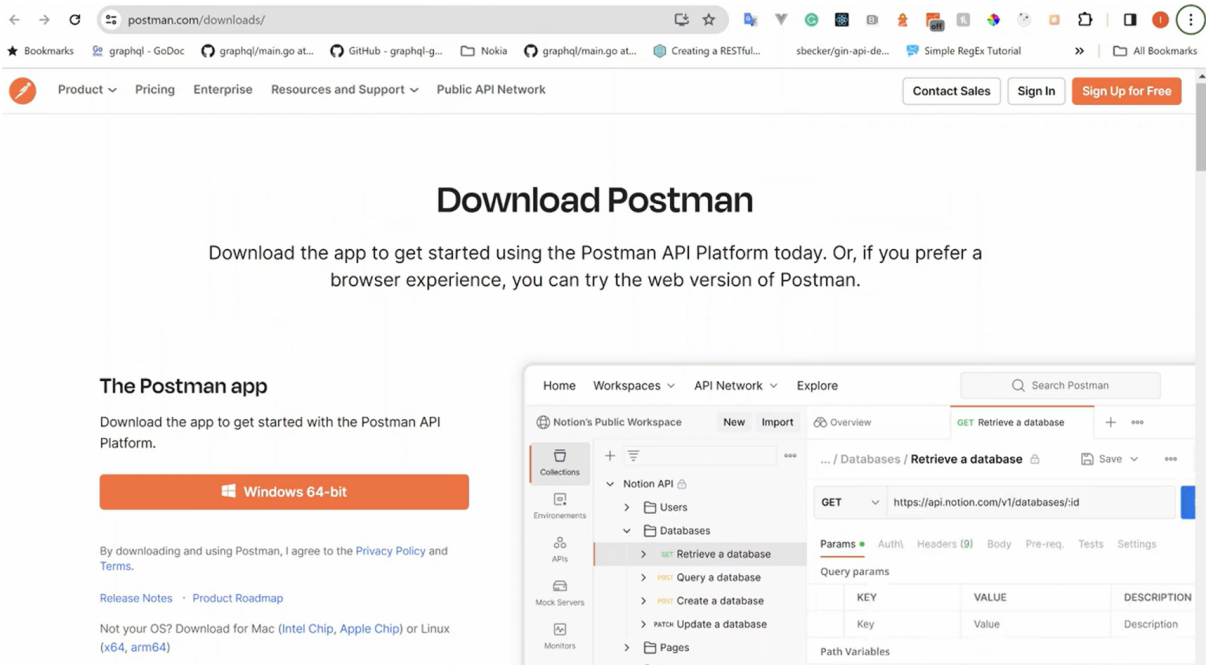
```
To view a single customer we GET it by specifying that customer's ID:

GET http://example.com/customers/123
Accept: application/json

Status Code: 200 (OK)
Content-type: application/json
```

PUT Request

```
We can update that customer by PUT ting the new data

PUT http://example.com/customers/123
Body:
{
    "name" = "Jone doe",
    "email" = "jone.doe@example.com"
}
A possible response header would have Status Code: 200 (OK),
to notify the client that the item with id 123 has been modified.
```

DELETE Request

```
We can delete the resource by an Id

Request:
DELETE http://example.com/customers/123

Response:
Status Code: 204 (NO CONTENT)
```

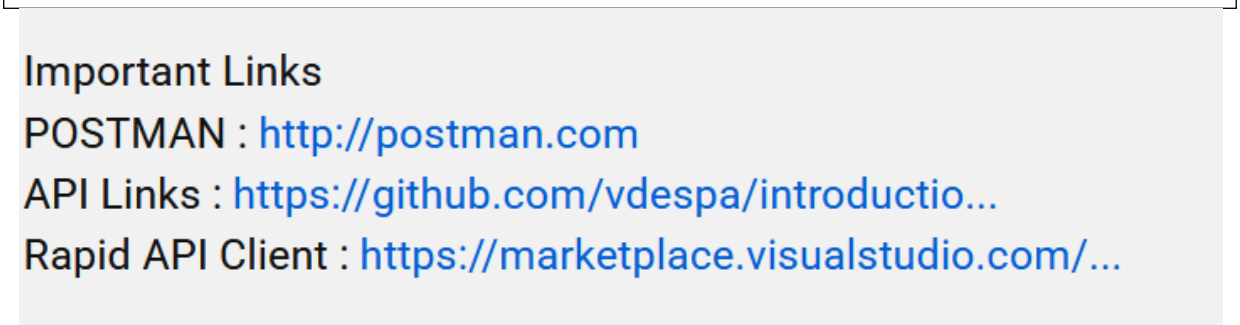


POSTman:

Link:

https://www.youtube.com/watch?v=CLG0ha_a0q8

https://github.com/vdespa/introduction-to-postman-course/blob/main/simple-books-api.md#endpoints



In postman check the following API

Status

GET `/status`

Returns the status of the API.

List of books

GET `/books`

Returns a list of books.

Optional query parameters:

- type: fiction or non-fiction
- limit: a number between 1 and 20.

Get a single book

GET `/books/:bookId`

Retrieve detailed information about a book.

Submit an order

POST `/orders`

Allows you to submit a new order. Requires authentication.

The request body needs to be in JSON format and include the following properties:

- `bookId` - Integer - Required
- `customerName` - String - Required

Example

```
POST /orders/
Authorization: Bearer <YOUR TOKEN>

{
  "bookId": 1,
  "customerName": "John"
}
```

The response body will contain the order Id.

Get all orders

GET `/orders`

Allows you to view all orders. Requires authentication.

Get an order

GET `/orders/:orderId`

Allows you to view an existing order. Requires authentication.

Update an order

PATCH `/orders/:orderId`

Update an existing order. Requires authentication.

The request body needs to be in JSON format and allows you to update the following properties:

- `customerName` - String

Example

```
PATCH /orders/PF6MflPDcuhWobZcgmJy5
Authorization: Bearer <YOUR TOKEN>

{
  "customerName": "John"
}
```

Delete an order

DELETE `/orders/:orderId`

Delete an existing order. Requires authentication.

The request body needs to be empty.

Example

```
DELETE /orders/PF6MflPDcuhWobZcgmJy5
Authorization: Bearer <YOUR TOKEN>
```

API Authentication

To submit or view an order, you need to register your API client.

POST `/api-clients/`

- **req.body**

- **req.body**
- **req.params**

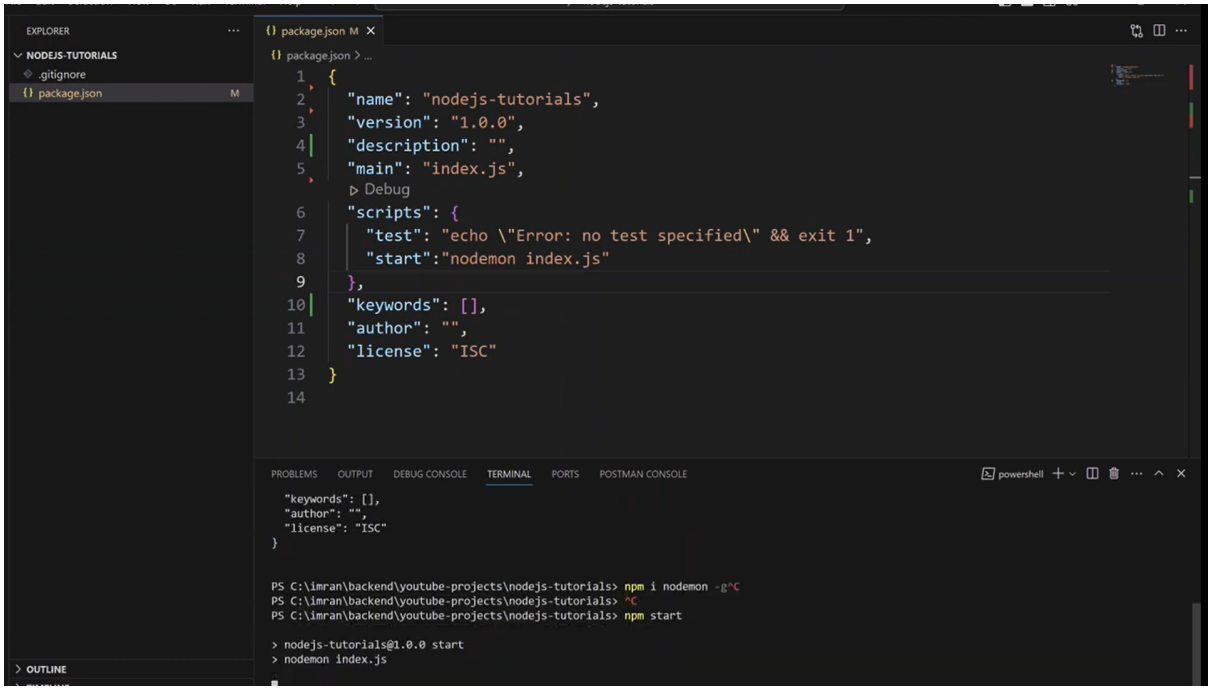
  /users/1

  /users/1/products/123

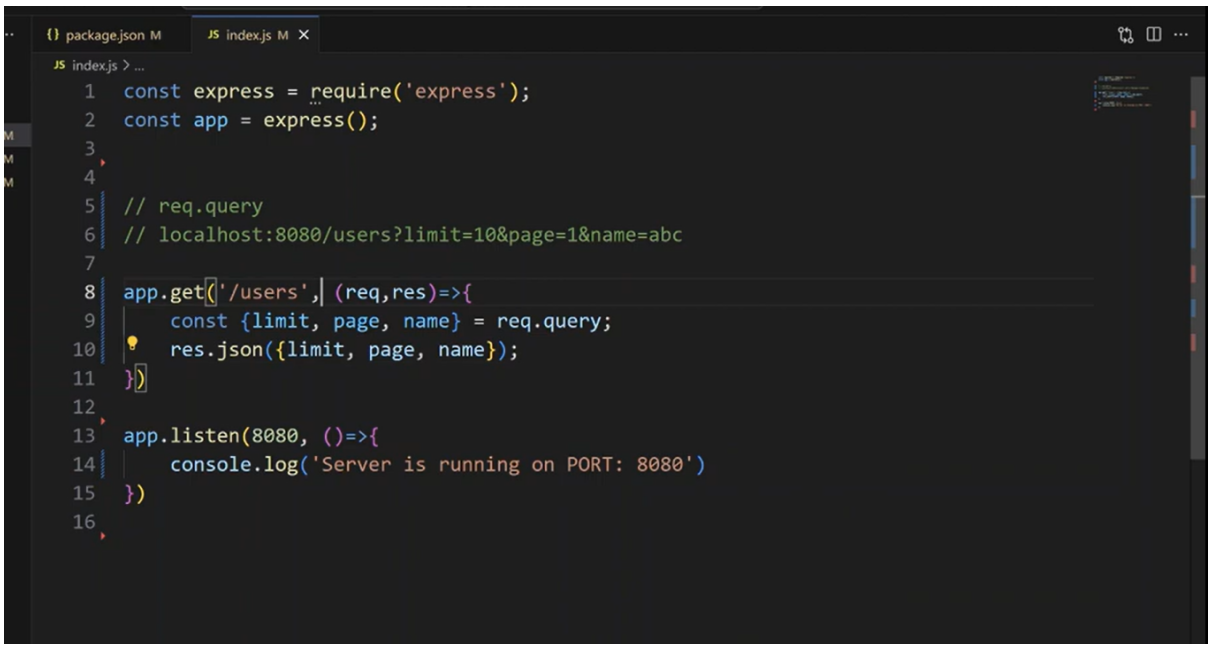
2

- **req.body**
- **req.params**
- **req.query**

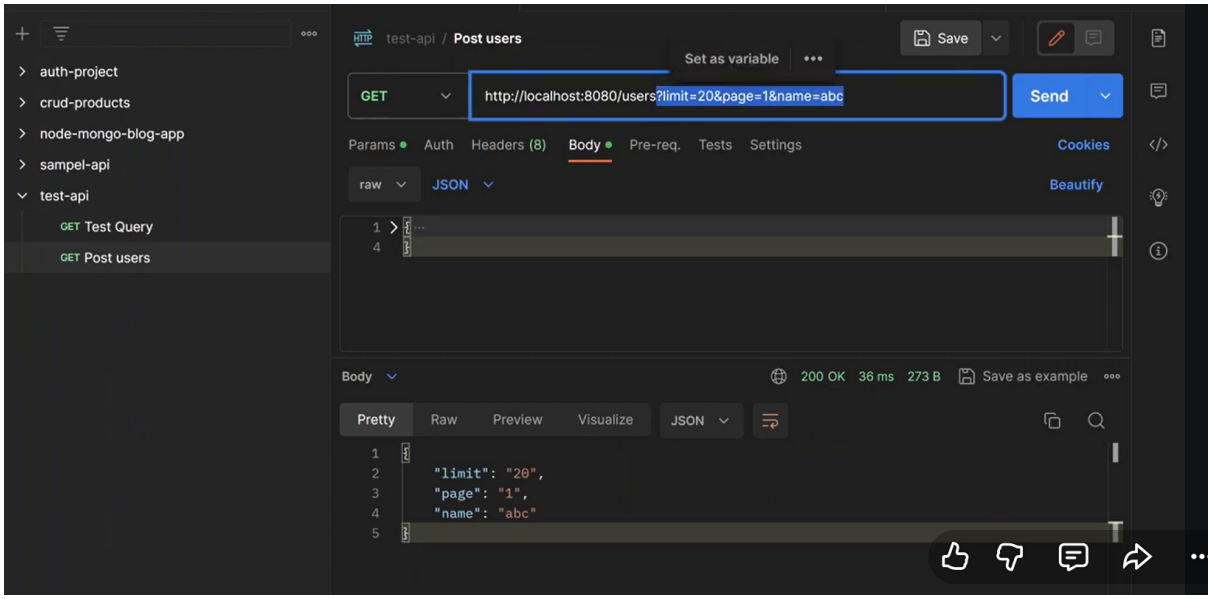
  http://localhost:8080?page=1&limit=10

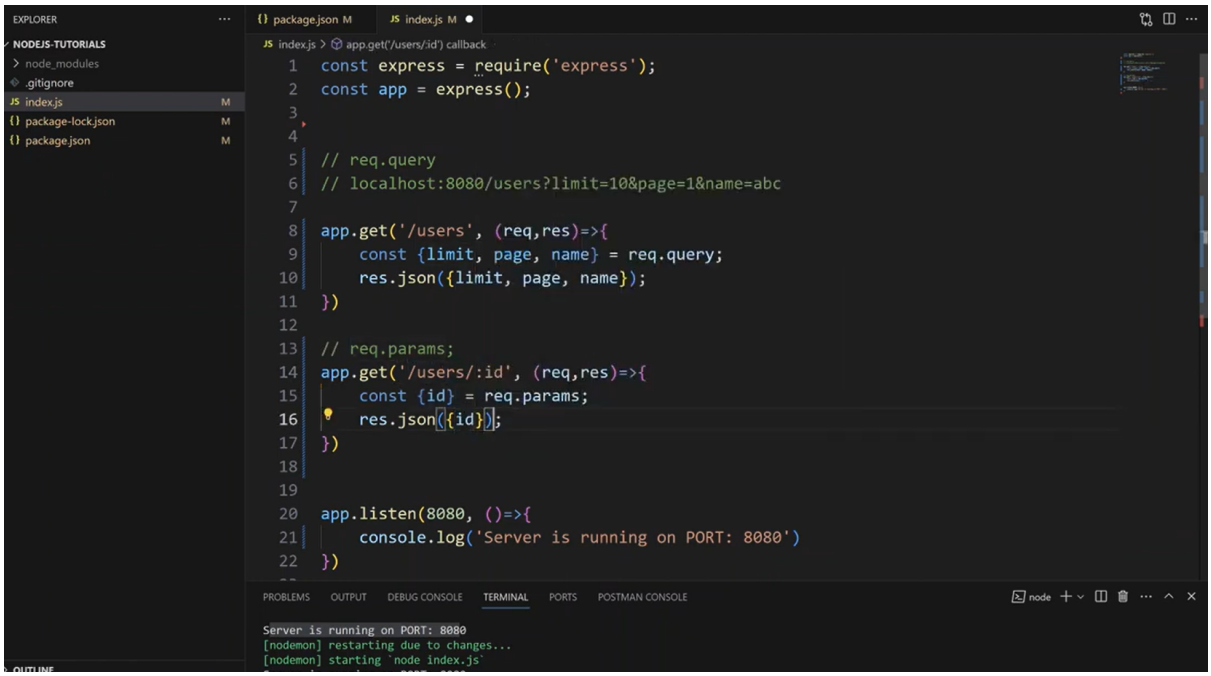


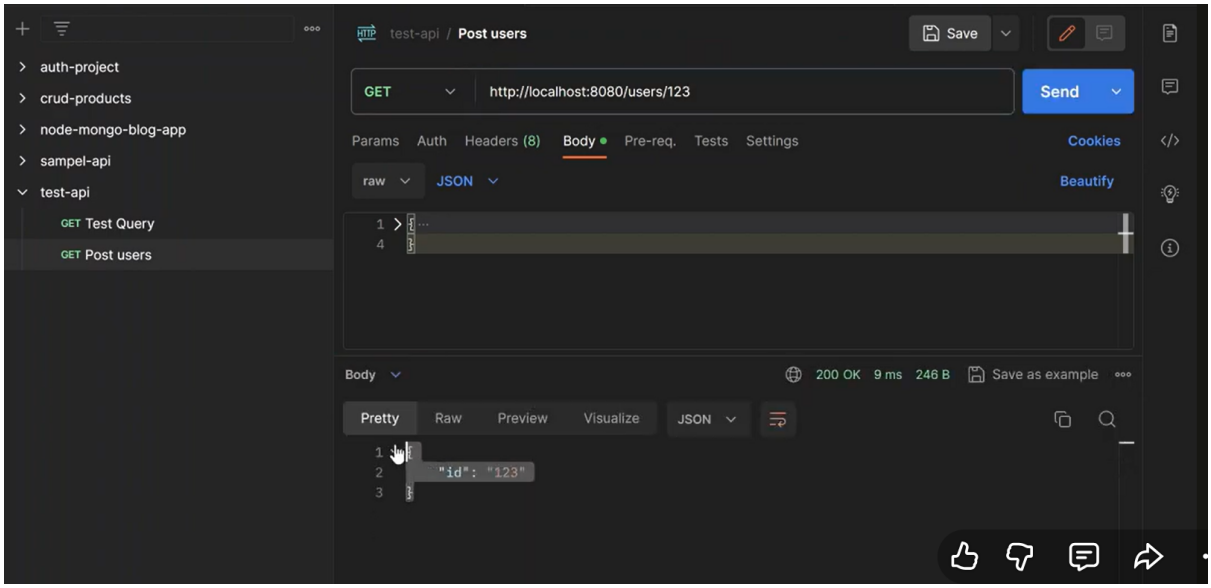
Index.js



On Postman


auth-project
crud-products
node-mongo-blog-app
sampel-api
test-api
GET Test Query
GET Post users
test-api / Post users
Save
Set as variable
GET
http://localhost:8080/users?limit=20&page=1&name=abc
Send
Params
Auth
Headers (8)
Body
Pre-req.
Tests
Settings
Cookies
raw
JSON
Beautify
Body
200 OK
36 ms
273 B
Save as example
Pretty
Raw
Preview
Visualize
JSON
{
"limit": "20",
"page": "1",
"name": "abc"
}


EXPLORER
NODEJS-TUTORIALS
node_modules
.gitignore
index.js
package-lock.json
package.json
package.json M
index.js M
index.js > app.get('/users/:id') callback
const express = require('express');
const app = express();
// req.query
// localhost:8080/users?limit=10&page=1&name=abc
app.get('/users', (req,res)=>{
const {limit, page, name} = req.query;
res.json({limit, page, name});
})
// req.params;
app.get('/users/:id', (req,res)=>{
const {id} = req.params;
res.json({id});
})
app.listen(8080, ()=>{
console.log('Server is running on PORT: 8080')
})
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
POSTMAN CONSOLE
node
Server is running on PORT: 8080
[nodemon] restarting due to changes...
[nodemon] starting `node index.js`



Express Framwork

- Introduction to Express
- Creating the Express Server
- Defining Routes
- Handling HTTP Request/Responses:
- Middleware
- CRUD Operation

Install express:

```
Is this OK? (yes)

C:\imran\backend\youtube-projects\nodejs-tutorials>
C:\imran\backend\youtube-projects\nodejs-tutorials>
C:\imran\backend\youtube-projects\nodejs-tutorials>npm i express body-parser

up to date, audited 98 packages in 2s

14 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\imran\backend\youtube-projects\nodejs-tutorials>npm i -D nodemon
```

Package.json file